DubzChain Phase 6.3 — Seed-Node Deployment Guide

This guide explains how to deploy, run, monitor, restart, and maintain a public DubzChain seed node.

Item	Recommended baseline
Node role	Stable public-facing relay or bootstrap node
P2P bind	0.0.0.0 on your chosen node port
RPC bind	127.0.0.1 unless public exposure is intentional
Advertise URL	ws://YOUR_PUBLIC_IP_OR_DOMAIN:PORT
Monitoring	/health, /status, /peers, /sync, /storage, /diagnostics, /metrics
Persistence	Keep chain data, wallet files, peer table, snapshot/checkpoint data

1. Purpose of the seed node

- A seed node is a stable public node that helps new peers discover and join the DubzChain network.
- It can also act as a bootstrap reference node for syncing, peer discovery, and basic network continuity.

2. Host and machine requirements

- Use a stable Linux VPS or server with a reliable internet connection.
- Choose a machine with enough disk space for your current storage mode, plus room for future growth.
- Use a Node.js version compatible with your TypeScript build and runtime setup.
- Keep the server clock accurate and the operating system updated.

3. Network ports and exposure

- Open the P2P port publicly so peers can connect to the node.
- Keep the RPC port bound to 127.0.0.1 unless you explicitly want to expose it.
- If exposing any public web UI or reverse proxy, keep P2P and RPC access rules intentional and documented.

4. Bind and advertise settings

- Use --host 0.0.0.0 for public P2P reachability.
- Use --rpc-host 127.0.0.1 unless public RPC access is intentional.

- Use --advertise ws://YOUR_PUBLIC_IP_OR_DOMAIN:PORT so peers learn the correct reachable node URL.
- Do not advertise localhost, LAN addresses, or stale public IP values.

5. Suggested startup examples

- Public bootstrap node: node dist/index.js 3001 --host 0.0.0.0 --rpc-host 127.0.0.1 --advertise ws://YOUR_PUBLIC_IP:3001 --automine --mine-empty
- Public relay/seed node: node dist/index.js 3001 --host 0.0.0.0 --rpc-host 127.0.0.1 --advertise ws://YOUR_PUBLIC_IP:3001
- Joining public peer: node dist/index.js 3002 --host 0.0.0.0 --rpc-host 127.0.0.1 --peer ws://YOUR_BOOTSTRAP_IP:3001 --advertise ws://YOUR_PUBLIC_IP:3002

6. Required files to preserve

- wallet.miner..json
- dubzchain..peers.json
- rpc.send-history..json if explorer send-history is used
- local chain data files
- snapshot/checkpoint files if enabled

7. Monitoring checklist

- Use /health for a quick up/down check.
- Use /status for height, tip, peer counts, sync status, and memory summary.
- Use /peers to confirm active connections.
- Use /sync to confirm catch-up and lag.
- Use /storage to confirm archival or pruned behavior.
- Use /diagnostics for deeper runtime and network inspection.
- Use /metrics if you want machine-readable monitoring output.

8. Restart procedure

- Stop the node cleanly.
- Do not delete chain data unless you are intentionally resetting the node.
- Restart using the same launch command and same advertised URL.
- After restart, verify /health, /status, /peers, and /debug/replay-verify.

9. Upgrade procedure

- Back up chain data and important files before upgrade.
- Confirm that network constants did not change unless you are doing an intentional reset or incompatible rollout.
- Rebuild the code, restart the process, and verify replay, peers, tip, and explorer access.
- Confirm wallet flow and send validation still work after the upgrade.

10. Abuse and incident handling

- If peers spam invalid traffic, inspect peer and diagnostics endpoints and review logs.
- If peer count spikes or sync stalls, verify the advertised URL, firewall, and network reachability.
- Keep RPC private by default to reduce attack surface.
- Use existing peer limits, bad-message handling, and network counters to understand node stress.

11. Final deployment checklist

- Public P2P port is open.
- RPC is private unless intentionally exposed.
- Advertise URL matches the real public endpoint.
- Chain data and peer files are persistent.
- Health, status, peers, sync, and replay checks pass.
- The node can restart and rejoin without manual repair.

Phase 6.3 completion note: once this guidance is saved and accepted as your deployment reference, you can mark 6.3 seed-node deployment guidance complete.